Digital Electronics Homework

2019 rev1

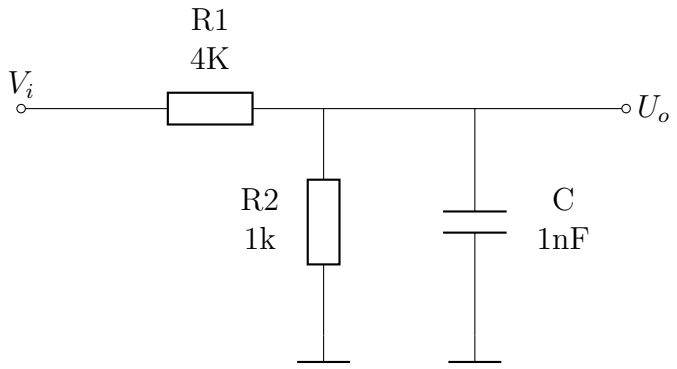
*** Deadline: May 20th, 10AM ***

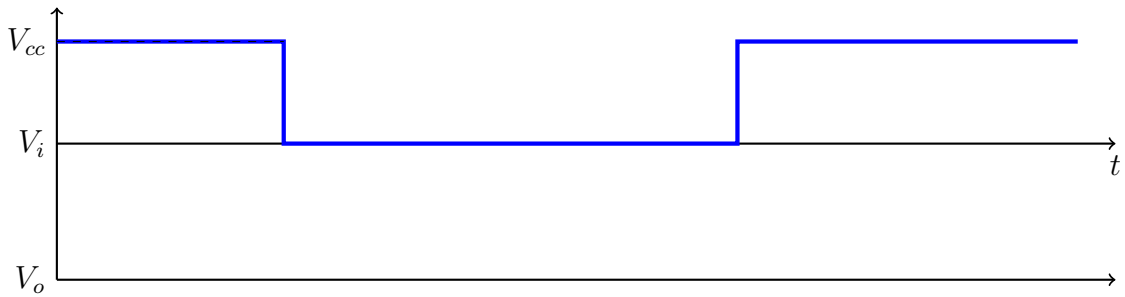
Solve the following problems. The solutions must be hand-written.

I advise you to read and analyze the problems independently before discussing them in groups. You must write the solutions independently – once you start writing the solution, refrain from any further discussion!! Do not distribute your solutions to other people and do not use other people's solutions. Please write down the name of all the people you collaborated with.

1 Problem 1

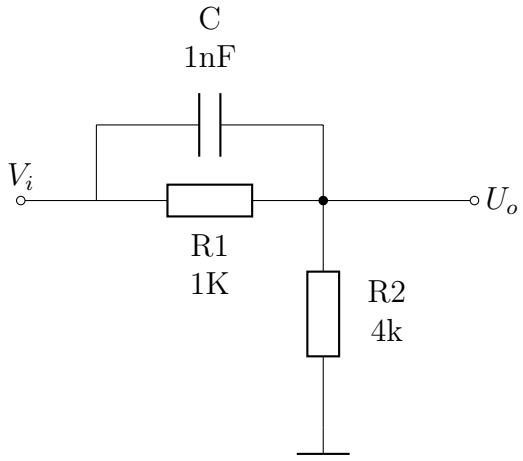
Plot the output signal for the following circuit and calculate the moments when the output signal crosses $0.1V_{cc}$ and $0.9V_{cc}$ (if they do), relative to the input signal edges. Assume $V_{cc} = 5V$. The duration of the input pulse is long enough for all the voltages to stabilize.

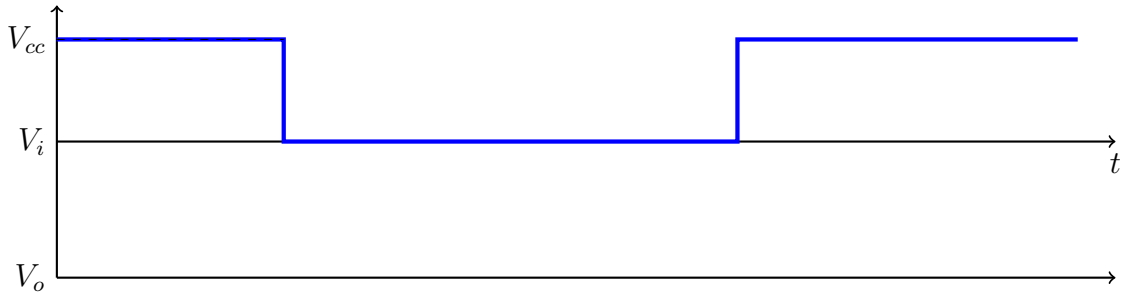




2 Problem 2

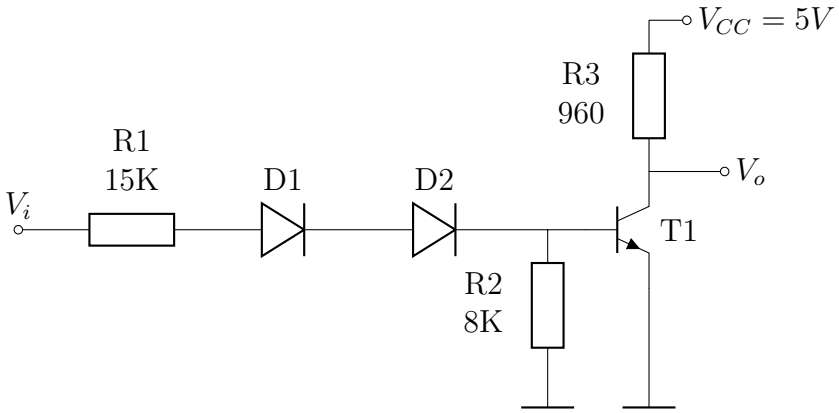
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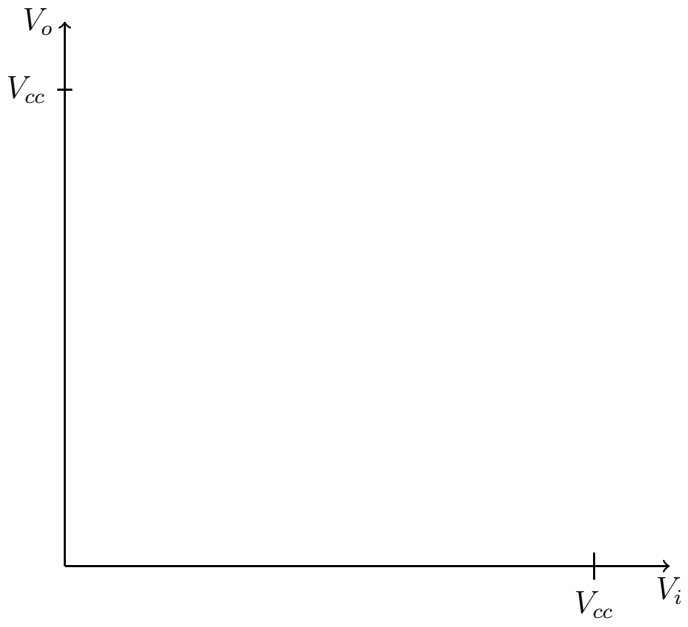




3 Problem 3

For the digital gate below, please determine the static dependency $V_o(V_i)$. Calculate V_{OL}, V_{IL}, V_{IH}, V_{OH}, MZL, MZH. Calculați factorul de supracomanda pentru T1. Se cunosc $U_D = 0.6V$, $U_{BE} = U_{BE0} = 0.8V$, $U_{CEsat} = 0.2V$, $\beta = 100$.

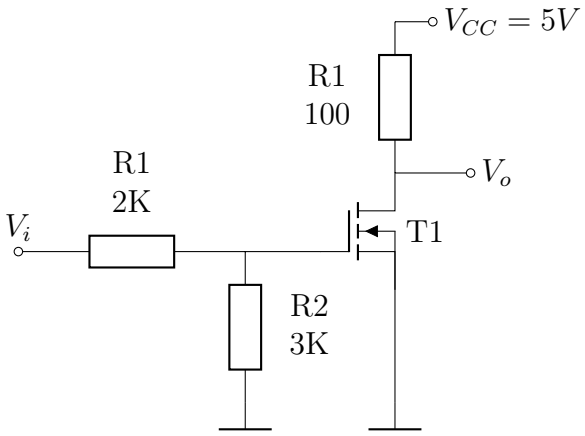




4 Problem 4

In the circuit below, assume the simplified Resistor-Saturation model for the MOSFET transistor, with $K = 0.1A/V^2$ and $V_T = 1V$.

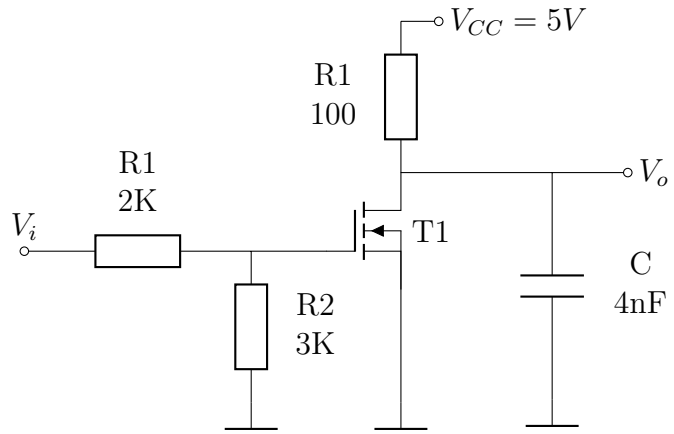
1. Calculate V_i such that the MOSFET is both in the linear region and in the saturation region
2. Calculate the V_o for the following cases:
   (a) $V_i = 2V$
   (b) $V_i = 5V$

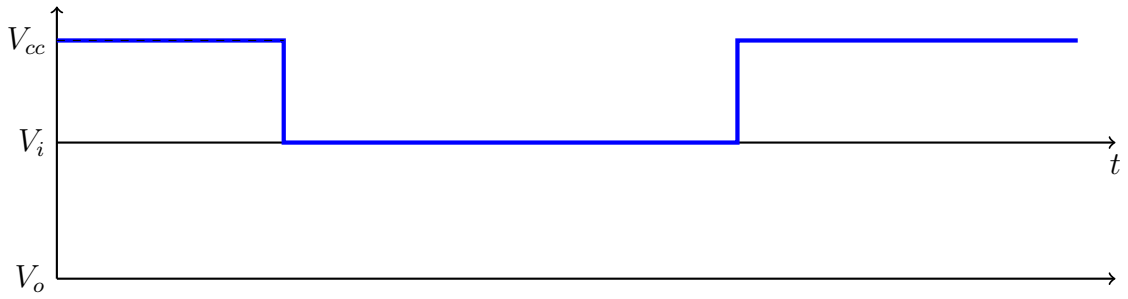


5 Problem 5

Plot the output signal for the following circuit and calculate the moments when the output signal crosses $0.1V_{cc}$ and $0.9V_{cc}$ (if they do), relative to the input signal edges. Assume $V_{cc} = 5V$. The duration of the input pulse is long enough for all the voltages to stabilize.

Assume the simplified Resistor-Saturation model for the MOSFET with $K = 0.1A/V^2$ and $V_T = 1V$. Ignore the input capacitance of the MOSFET.





6 Problem 6

Design a CMOS circuit which implements $\overline{A}B \vee C$. All variable represent digital input signals. The only components you are allowed to use are MOS transistors. Try to minimize the number of transistors used. Explain briefly your design and how it works.